$\ulcorner x^p - a.$

Let F be a field of characteristic $p \in \mathbb{N}$, and $a \in F^\times$.
If $x^p - a$ has no root in F, then $x^p - a$ is irreducible over F.

Proof. Let $\alpha \in \overline{F}$ such that $\alpha^p - a = 0$. Then, $x^p - a$ splits over $\overline{F}$,

$$x^p - a = x^p - \alpha^p = (x-\alpha)^p$$

because $ch(F) = p$ implies; by the binomial theorem,

$$(x-\alpha)^p = \sum_{n=0}^{p} x^n \cdot \alpha^{p-n} \cdot \binom{p}{p-n} = x^p - \alpha^p.$$

Therefore, if $x^p - a$ factors by $x^p - a = f(x) \cdot g(x)$ where $\deg f, \deg g < p$
with $f, g \in F[x]$, then over $\overline{F}$, by the unique factorization,

$$f(x) = (x-\alpha)^{p-k}, \quad g(x) = (x-\alpha)^k \quad \text{for some } 1 \le k \le p-1,$$

But, $g(x) = (x-\alpha)^k = x^k - k\alpha x^{k-1} \cdots + (-\alpha)^k \in F[x]$ implies $k\alpha \in F$.
Since p is prime, $\gcd(k, p) = 1$, so $nk\alpha = (1+p)\alpha = \alpha$ for some $n \in \mathbb{N}$,
it contradicts to $\alpha \notin F$.
(or, Since F is a field, $\alpha = (k \cdot 1_F)^{-1} \cdot (k \cdot 1_F)\alpha$).